

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 8_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Write a program to validate the email address and display suitable exceptions if there is any mistake.

Create 3 custom exception classes as below

DotException AtTheRateException DomainException

A typical email address should have a "." character, and a "@" character, and also the domain name should be valid. Valid domain names for practice be 'in', 'com', 'net', or 'biz'.

Display Invalid Dot usage, Invalid @ usage, or Invalid Domain message based on email id.

Get the email address from the user, validate the email by checking the

above-mentioned criteria, and print the validity status of the input email address.

Input Format

The first line of input contains the email to be validated.

Output Format

The output prints a Valid email address or an Invalid email address along with the suitable exception

If email ends with . or contains not exactly one . after @, it throws:

DotException: Invalid Dot usage

Invalid email address

If @ appears not exactly once, it throws:

AtTheRateException: Invalid @ usage

Invalid email address

If the part after the last dot is not among accepted domains:

DomainException: Invalid Domain

Invalid email address

If all conditions satisfied then print:

Valid email address

Refer to the sample input and output for format specifications.

Sample Test Case

Input: sample@gmail.com

Output: Valid email address

Answer

```
// You are using Java
import java.util.Scanner;
class DotException extends Exception
{
    public DotException(String s)
    {
        super(s);
    }
}
class AtTheRateException extends Exception
{
    public AtTheRateException(String s)
    {
        super(s);
    }
}
class DomainException extends Exception
{
    public DomainException(String s)
    {
        super(s);
    }
}
class Main
{
    public static int a=0,b=0,c=0;
    public static void dot(String s) throws DotException
    {
        String[] p=s.split("@");
        String s1=p[1];
        int v=0;
        for(int i=0;i<s1.length();i++)
```

```

{
    if(s1.charAt(i)=='.')
    {
        v++;
    }
}

if(v!=1 || s.endsWith("."))
{
    throw new DotException("Dot usage\nInvalid email address");
}

a=1;
}

public static void rate(String s) throws AtTheRateException
{
    int c=0;
    for(int i=0;i<s.length();i++)
    {
        if(s.charAt(i)=='@')
        {
            c++;
        }
    }

    if(c!=1)
    {
        throw new AtTheRateException("@ usage\nInvalid email address");
    }

    b=1;
}

public static void domain(String s) throws DomainException
{
    if(!s.endsWith("in") && !s.endsWith("com") && !s.endsWith("net") && !
s.endsWith("biz"))
    {
        throw new DomainException("Domain\nInvalid email address");
    }

    c=1;
}

public static void main(String[] args)
{
    Scanner sc=new Scanner(System.in);
    String s=sc.nextLine();
}

```

```
try
{
    rate(s);
    dot(s);
    domain(s);
    if(a==1 && b==1 && c==1)
    {
        System.out.print("Valid email address");
    }
}

catch(DotException e)
{
    System.out.print("DotException: Invalid "+e.getMessage());
}

catch(AtTheRateException e)
{
    System.out.print("AtTheRateException: Invalid "+e.getMessage());
}

catch(DomainException e)
{
    System.out.print("DomainException: Invalid "+e.getMessage());
}

}
```

Status : Correct

Marks : 10/10